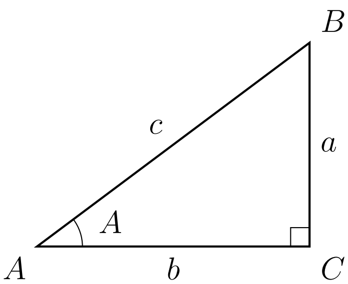
Solving Right Triangles: Elevation and Depression

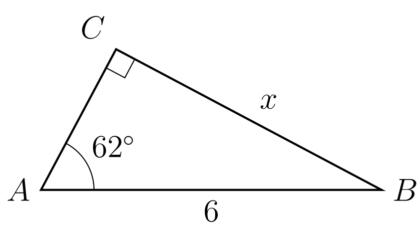
Directions

- An **angle of elevation** is measured up from a horizontal line; an **angle of depression** is measured down from one. Because the two horizontal lines are parallel, the angle of depression from the top equals the angle of elevation from the bottom.
- Work in degree mode. Round every length to two decimal places and every angle to the nearest tenth of a degree.
- Each problem carries its own diagram drawn to scale from that problem's givens. Write the unit on every answer.



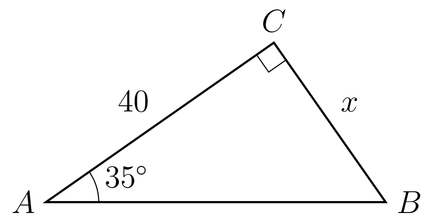
How every triangle here is labelled. No values shown: use the numbers given in each problem.

1. A 6 m ladder leans against a wall and makes an angle of 62° with the level ground. How far up the wall does the top of the ladder reach, in metres?



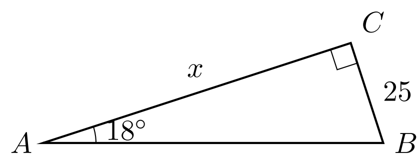
Answer: ________________ m

2. A surveyor stands 40 ft from the foot of a flagpole on level ground. The angle of elevation from the surveyor's eye to the top of the pole is 35°. Find the height of the pole above eye level, in feet.



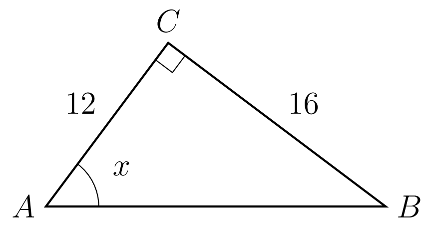
Answer: __________________ ft

3. From the top of a 25 m lighthouse the angle of depression to a boat is 18°. Find the horizontal distance from the foot of the lighthouse to the boat, in metres.



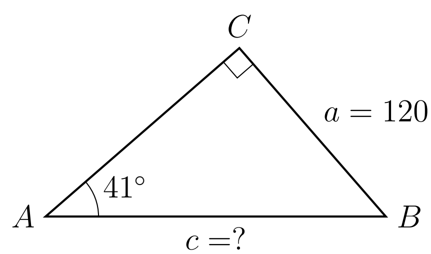
Answer: __________________ m

4. A guy wire runs from an anchor on level ground 12 m from the base of a mast up to a point 16 m above the ground on the mast. Find x, the angle of elevation of the wire from the anchor, to the nearest tenth of a degree.



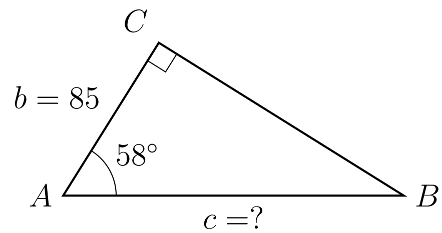
Answer: ________________ degrees

5. A drone hovers 120 m above level ground. The angle of depression from the drone to a landing pad on the ground is 41°. Find the straight-line distance from the drone to the pad, in metres.



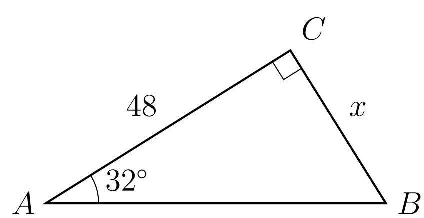
Answer: ________________ m

6. A hiker stands 85 ft from the base of a vertical cliff. The angle of elevation to the top of the cliff is 58°. Find the length of the hiker's line of sight to the top of the cliff, in feet.



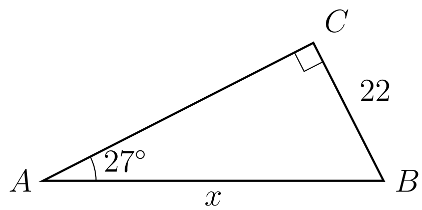
Answer: ________________ ft

7. From a window 15 m above level ground, the angle of elevation to the top of a tower is 32°. The tower stands 48 m away horizontally. Find the total height of the tower above the ground, in metres.



Answer: ________________ m

8. A zip line runs from a platform 22 m above level ground down to a landing point on the ground. Along the zip line the angle of depression is 27°.



(a) Find the length of the zip line, in metres.

Answer: ________________ m

(b) Find the horizontal distance from the base of the platform to the landing point, in metres.

Answer: ________________ m

(c) Explain why the angle of depression measured at the platform is equal to the angle of elevation measured at the landing point.